Higgs Boson Signal Strength in the Four-Lepton Final State with CMS Open Data

Local analysis of derived CMS Open Data ntuples

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Abstract

This note presents a simplified detector-level measurement of the Higgs boson signal strength in the $H \to ZZ^* \to 4\ell$ final state using 10.0 fb^{-1} of flat ntuples derived from CMS Open Data NanoAOD. The observed signal-strength modifier is

$$\mu = 1.647^{+0.636}_{-0.522}.$$

The fitted mass parameter is

$$m_H = 128.628^{+0.766}_{-0.678}\,\text{GeV}.$$

1 Inputs and Reconstruction

The input files contain a flat `h4lTree` produced by `h4l_ntuplize.py` from CMS Open Data NanoAOD. The upstream NanoAOD format stores event information in ROOT TTrees and can be analyzed directly with Python tools such as uproot and awkward. The ntuplizer builds the best same-flavour opposite-sign four-lepton candidate and stores reconstructed candidate variables, lepton IDs, isolation, trigger masks, and metadata. The analysis uses 10.0 fb^{-1} at $\sqrt{s} = 13$ TeV. Signal templates combine ggH, VBF, WH, ZH production, the irreducible background combines $q\bar{q} \to ZZ$ and $gg \to ZZ$, and the reducible background is modeled with DY+jets and `ttbar` simulation.

Input	Treatment	Notes
Data luminosity	Fixed	10.0 fb^{-1} from input metadata
Signal normalization	Fixed SM prediction	Cross sections from the prompt
Irreducible background	MC templates and sideband model	qqZZ and ggZZ components
Reducible background	MC estimate	DY+jets and `ttbar` only

Table 1: Summary of the main inputs and their treatment in the fit.

2 Selection

The nominal selection requires medium PF muons, WP90-or-cutBased electrons, relative isolation below 0.40, $|\text{SIP3D}| < 5$, leading and subleading lepton transverse momenta above 20 and 10 GeV, $40 < m_{Z1} < 120$ GeV, $12 < m_{Z2} < 120$ GeV, and a nonzero trigger bit mask. The selected four-lepton mass distribution is shown in Fig. 1.

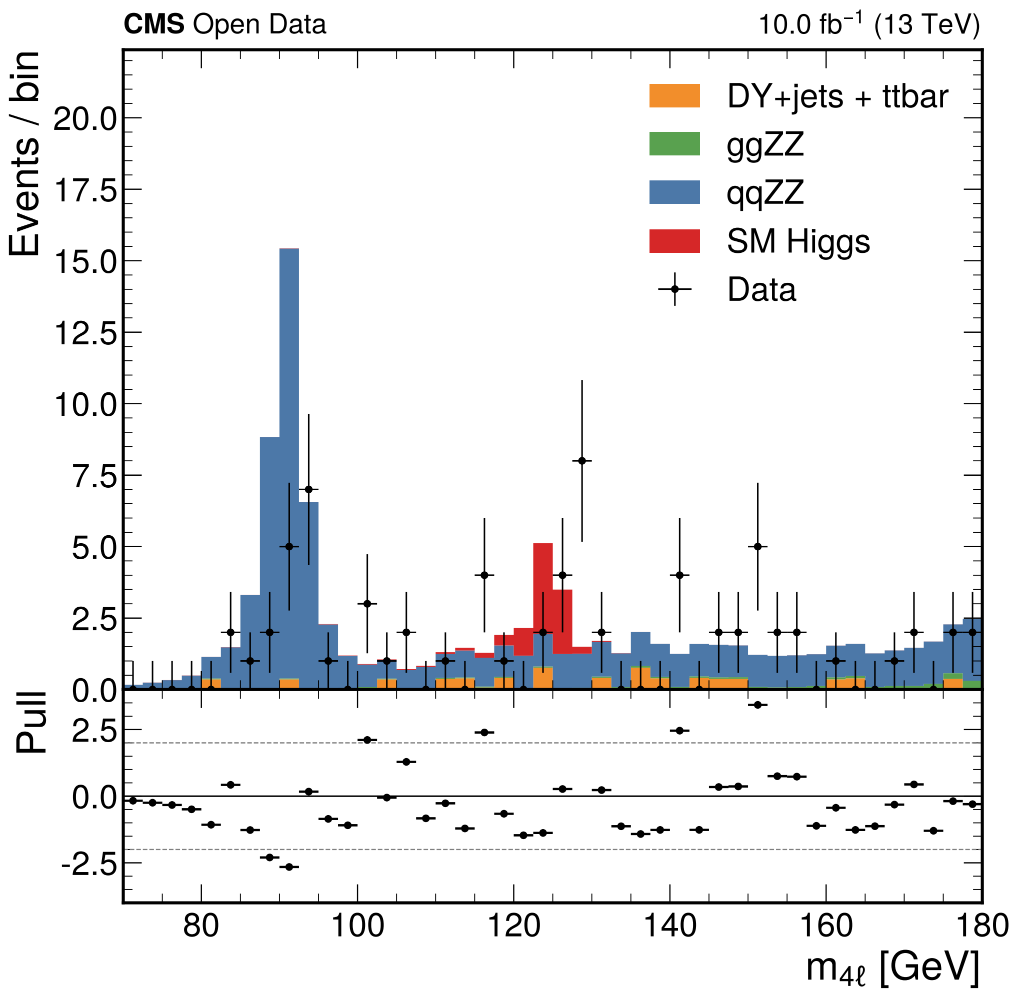

Figure 1: Selected four-lepton mass spectrum with SM signal and backgrounds.

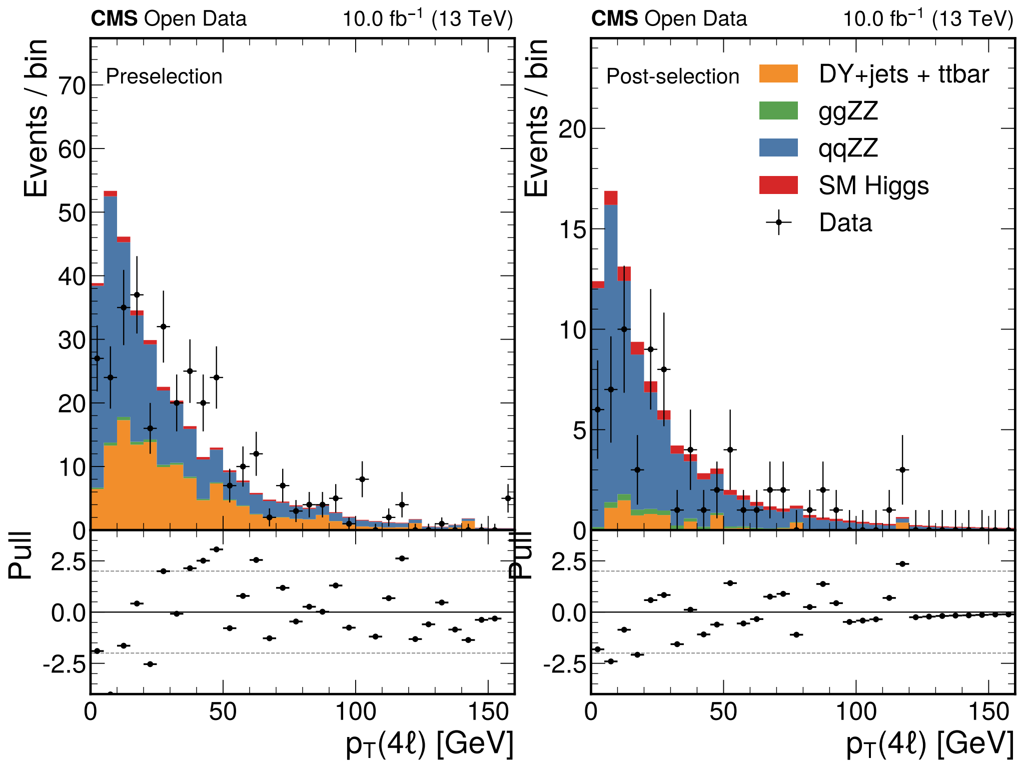

Figure 2: Four-lepton transverse momentum before and after the nominal selection.

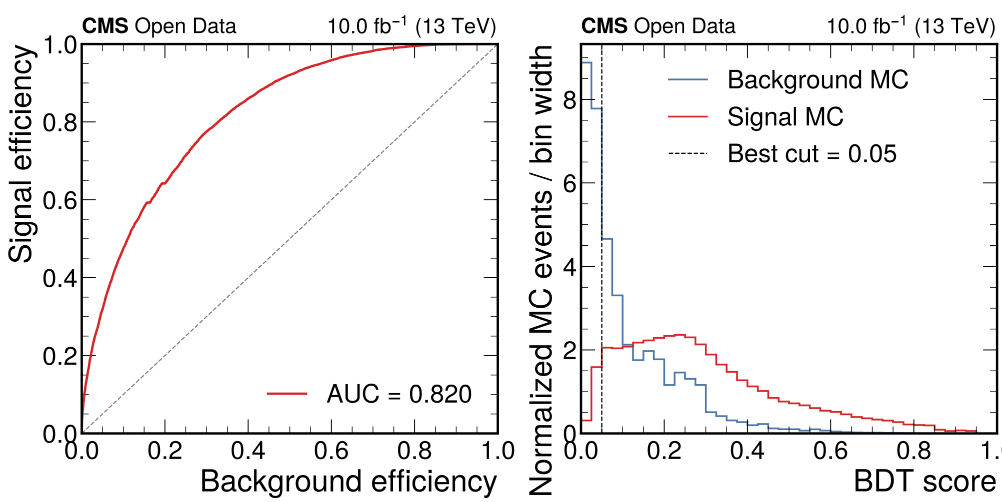

Figure 3: MC-only BDT cross-check performance in the fit mass range.

The BDT is used only as a validation cross-check, not in the nominal signal strength fit. It is trained on non-mass kinematic and object-quality variables in the $105 < m_{4\ell} < 160$ GeV fit range. The test AUC is 0.820; the best scan point is a score threshold of 0.05, with an expected Higgs-window sensitivity of 1.979. The BDT-score data/MC agreement is shown in Fig. 4 and gives $\chi^2/\mathrm{ndf} = 0.927$ with $p = 0.598$, pull RMS 0.950, and maximum absolute pull 3.620.

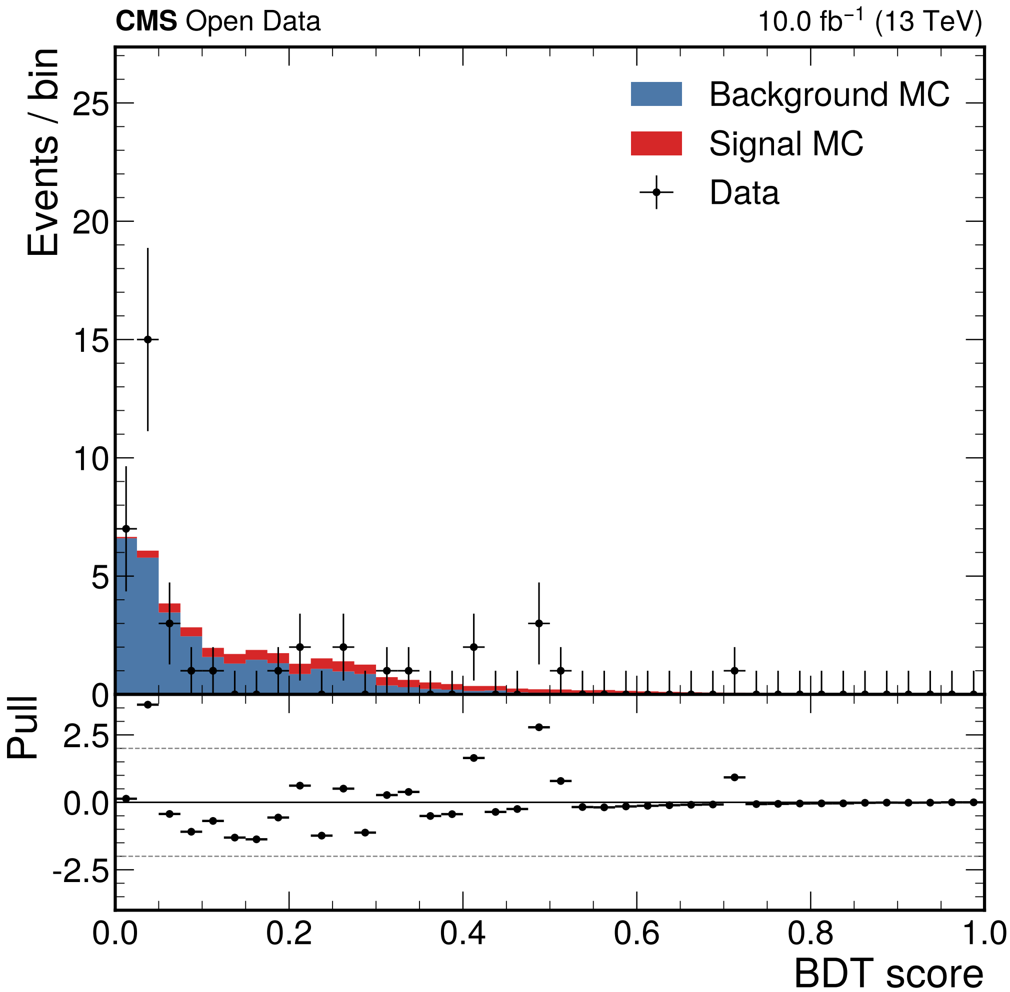

Figure 4: Data/MC comparison of the BDT score in the fit mass range.

3 Inference

The fit is a binned likelihood in $105 < m_{4\ell} < 160$ GeV with 2.5 GeV bins. The model is $\lambda_i = \mu S_i(m_H) + \beta f_i(k)$, where $S_i(m_H)$ is the selected SM Higgs MC template shifted by a fitted mass offset and $f_i(k)$ is an exponential sideband-background model. The background normalization has a 50% Gaussian prior around the total MC background prediction. The expected Asimov fit gives $\mu = 1.000^{+0.572}_{-0.450}$.

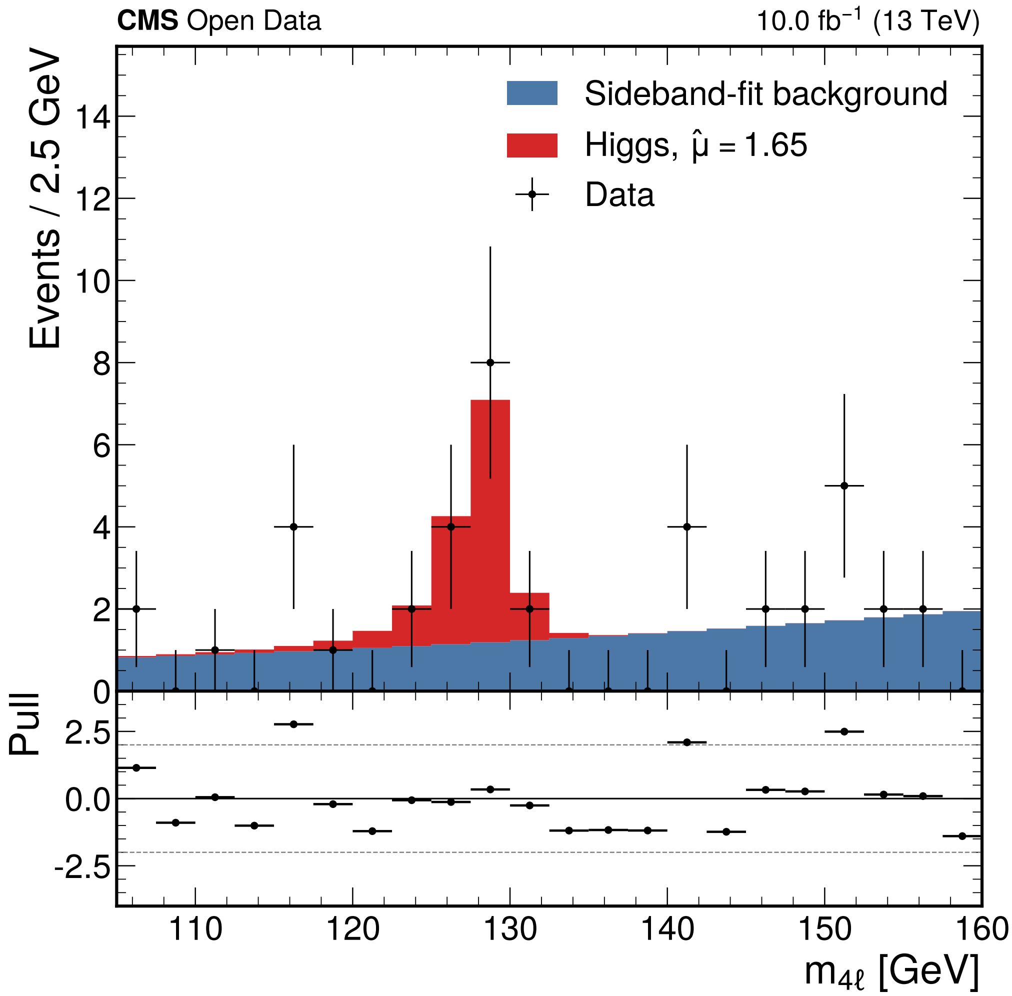

Figure 5: Observed post-fit four-lepton mass spectrum.

Nuisance	Fit value	Prior width	Pull
Combined background	28.80	14.86	-0.06
qqZZ normalization	1.139	0.15	0.92
ggZZ normalization	1.026	0.30	0.09
Reducible normalization	0.526	0.50	-0.95

Table 2: Nominal combined-background nuisance and diagnostic component-normalization pulls.

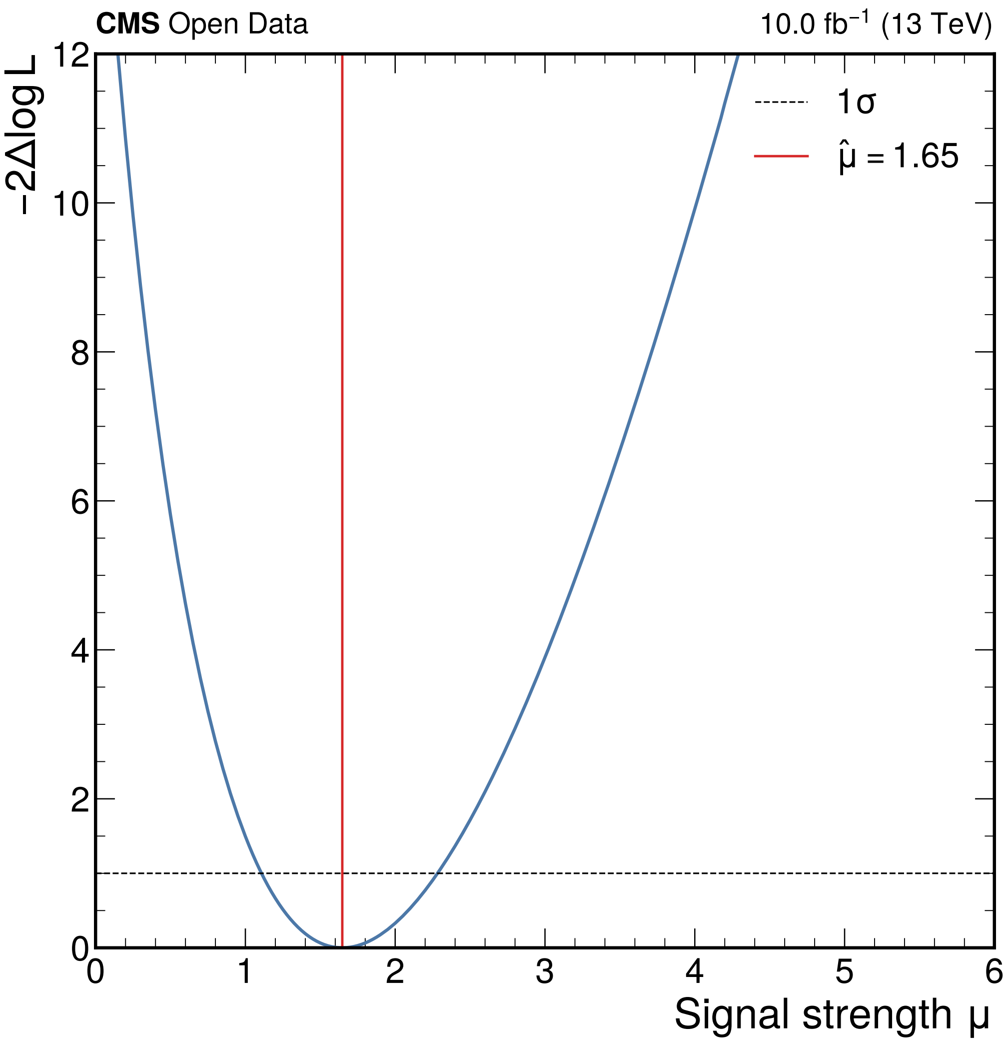

Figure 6: Profile likelihood scan for the signal-strength modifier.

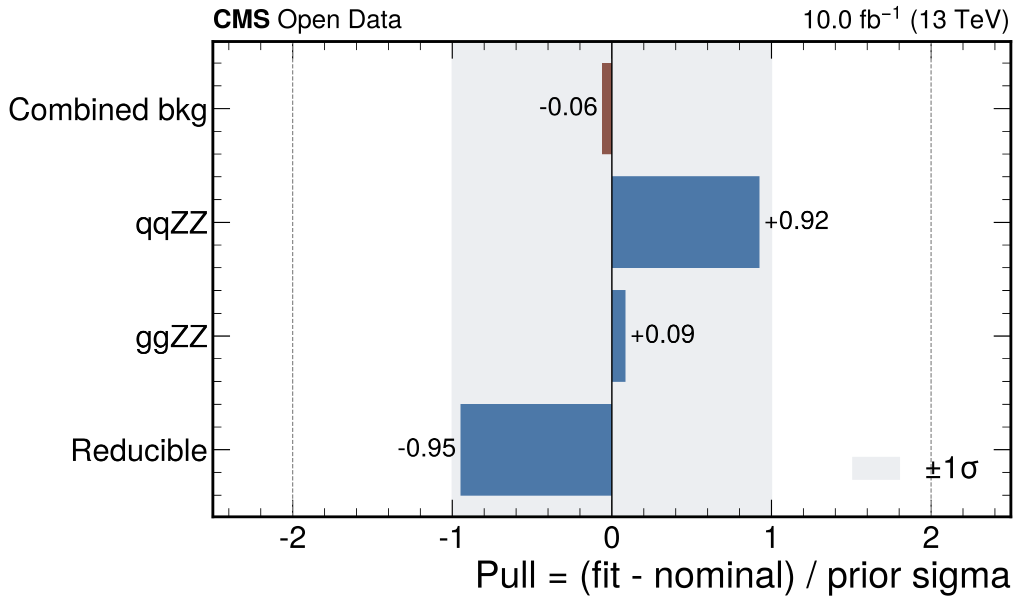

Figure 7: Background-normalization nuisance pulls. Component pulls are shown as a diagnostic cross-check; the nominal fit uses the combined-background constraint.

4 Results

The observed result is

$$\mu = 1.647^{+0.636}_{-0.522},$$

and

$$m_H = 128.628^{+0.766}_{-0.678}\,\text{GeV},$$

with a goodness of fit $\chi^2/\text{ndf} = 1.738$ and $p = 0.027$. The simplified background-only test statistic corresponds to 4.00 standard deviations.

Channel	Data	Signal expected	Background expected
4mu	3	2.49	1.87
4e	3	1.14	1.15
2e2mu	8	3.00	2.62

Table 3: Yields in the 120–130 GeV Higgs window.

5 Limitations

The result should be interpreted as a reproducible Open Data exercise rather than a full CMS publication-quality measurement. The flat ntuple does not contain all correction payloads needed for lepton efficiency, energy scale, and resolution systematics, and the reducible background is modeled with DY+jets and `ttbar` MC rather than a fake-rate method, as requested. The explicit list of unused nuisance parameters is written to the Phase 4 nuisance-summary artifact.

References

[1] CMS Collaboration, Measurements of properties of the Higgs boson decaying into the four-lepton final state in pp collisions at $\sqrt{s} = 13$ TeV, JHEP 11 (2017) 047, CMS-HIG-16-041.

[2] CERN Open Data Portal, Getting Started with CMS NanoAOD Open Data, `https://opendata.cern.ch/docs/cms-getting-started-nanoaod`.